

VAL MARIE SOUTHEAST, SK, Canada

WMO: 711370

Lat: 49.060N

Lon: 107.590W

Elev: 796

StdP: 92.12

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.3	-29.5	-37.4	0.1	-33.2	-33.2	0.2	-29.4	14.3	-3.2	13.1	-3.6	1.5	60	0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	33.5	17.0	31.4	16.5	29.3	16.0	19.4	27.9	18.2	27.0	17.3	26.0	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	13.1	22.3	15.3	12.0	20.9	14.2	11.1	20.0	58.7	27.8	54.8	27.1	51.6	26.0	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.5	11.0	9.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-38.6	36.6	3.8	1.8	-41.3	38.0	-43.5	39.0	-45.6	40.1	-48.4	41.4		
			-38.6	21.6	3.7	1.6	-41.3	22.7	-43.5	23.6	-45.5	24.5	-48.3	25.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-12.0	-10.1	-3.2	4.8	10.6	15.4	19.5	18.7	12.8	5.1	-3.4	-10.6		
	DBStd	12.70	8.81	8.41	7.98	5.24	4.50	3.59	3.35	3.67	4.55	5.02	7.19	8.13		
	HDD10.0	3097	681	562	410	166	48	3	0	1	23	164	402	638		
	HDD18.3	5352	939	795	668	404	242	101	26	42	174	411	652	897		
	CDD10.0	923	0	0	1	11	66	165	294	268	106	12	1	0		
	CDD18.3	134	0	0	0	0	2	13	62	51	7	0	0	0		
	CDH23.3	2721	0	0	0	9	97	283	1073	983	263	12	0	0		
	CDH26.7	1094	0	0	0	1	20	82	462	434	94	1	0	0		
(14) Wind	WSAvg	4.1	3.6	3.6	4.2	4.8	4.9	4.4	3.8	3.9	4.0	4.1	3.8	3.5		
(15) Precipitation	PrecAvg	389	15	6	20	35	76	69	60	37	31	19	13	14		
	PrecMax	589	41	13	31	71	147	159	146	111	91	124	38	86		
	PrecMin	268	5	0	8	10	33	22	19	1	2	2	0	0		
	PrecStd	86	10	4	6	17	33	37	35	29	21	25	10	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	10.1	18.0	24.6	29.0	32.1	36.4	36.1	32.5	25.2	16.7	8.0		
		MCWB	4.1	4.9	9.1	11.4	14.2	16.8	17.6	17.1	15.6	12.6	8.7	3.2		
	2%	DB	4.4	5.7	13.7	20.8	26.1	29.0	33.8	33.6	29.6	20.4	12.2	4.6		
		MCWB	1.8	2.1	6.5	10.3	13.7	16.3	17.5	16.5	14.6	10.9	6.2	1.5		
	5%	DB	2.6	3.2	10.4	17.7	23.5	26.6	31.6	31.3	26.5	17.0	8.8	2.5		
		MCWB	0.9	1.0	4.8	8.5	12.5	15.6	17.3	16.0	13.8	9.0	4.1	0.4		
	10%	DB	0.5	1.0	7.2	14.9	20.6	24.3	29.3	29.0	23.2	14.1	5.9	0.5		
		MCWB	-0.8	-0.3	3.0	7.2	11.2	14.6	16.9	15.6	12.6	7.7	2.5	-1.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	5.2	9.2	12.6	16.1	20.2	21.5	20.0	17.5	13.6	9.1	3.7		
		MCDB	6.9	9.6	16.8	22.4	25.1	27.6	29.1	29.1	27.6	22.8	15.7	7.2		
	2%	WB	2.0	2.5	7.0	10.7	14.4	18.0	19.9	18.5	15.7	11.4	6.7	1.9		
		MCDB	4.0	5.2	12.9	19.0	23.6	25.1	28.1	27.5	25.4	19.0	11.6	4.4		
	5%	WB	0.9	1.0	5.1	9.3	13.2	16.6	18.8	17.5	14.5	9.6	4.5	0.5		
		MCDB	2.5	3.0	9.7	16.7	21.4	23.7	27.7	26.9	24.0	15.8	8.1	2.3		
	10%	WB	-0.8	-0.3	3.3	7.7	12.1	15.6	17.8	16.6	13.3	8.2	2.7	-1.0		
		MCDB	0.6	1.1	6.9	14.0	19.0	22.4	26.5	26.1	22.2	13.5	5.7	0.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	12.3	12.2	13.9	14.7	14.1	16.5	17.1	16.2	14.2	12.4	12.0		
		MCDBR	12.6	13.4	16.7	19.8	19.9	18.8	20.5	21.5	22.7	20.0	16.4	13.4		
		MCWBR	10.4	10.3	10.2	10.6	9.5	8.5	7.9	8.2	10.3	11.1	10.8	10.7		
		MCDBR	12.4	13.0	15.8	17.9	16.8	15.6	17.0	17.2	19.4	18.2	15.8	13.0		
	5% WB	MCWBR	10.3	10.3	10.0	9.9	8.4	8.1	7.6	7.2	9.2	10.5	10.7	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.252	0.275	0.327	0.348	0.351	0.359	0.373	0.326	0.285	0.257	0.239		
	taud		2.462	2.520	2.533	2.386	2.372	2.420	2.409	2.340	2.440	2.545	2.523	2.477		
	Ebn at Noon		847	929	954	922	911	908	895	863	877	871	828	811		
	Edh at Noon		59	71	84	109	116	111	111	113	92	70	56	51		
(44) All-Sky Solar Radiation	RadAvg		1.33	2.33	3.61	4.92	5.88	6.37	6.89	5.66	4.10	2.58	1.49	1.05		
	RadStd		0.11	0.14	0.19	0.27	0.42	0.40	0.39	0.40	0.34	0.26	0.15	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page